

the availability of a newer Software Image in response to a [QueryImage](#) command. This algorithm will be called the "OTA Image Selection Logic" thereafter.

The OTA Image Selection Logic MAY use any data it deems useful, either local to the equipment or Node hosting the OTA Provider, or remote through external networks, to determine whether an updated Software Image is available (see [Section 11.20.3.3, "Availability of Software Images"](#)).

OTA Provider requests are idempotent. In cases where an OTA Requestor is repeating a request it has already done, and the OTA Provider can detect this, it SHALL NOT behave differently on any subsequent attempt compared to the first, unless a new Software Image has become available in the meantime. That is, an OTA Provider SHALL NOT prevent an OTA Requestor from trying to make the same query more than once. This requirement is critical to ensure that OTA Requestors which encounter error conditions during OTA Image Query or OTA Image Transfer can eventually succeed through retrying the same operation more than once.

Upon final determination of the outcome of the [QueryImage](#) command, the OTA Provider SHALL reply with a [QueryImageResponse](#) command.

If an OTA Provider employs synchronous proxying (e.g. proxy-while-downloading) method to reach off-Fabric Software Images and provide them over [BDX Protocol](#) to OTA Requestors, it SHALL respond with a Status of DownloadProtocolNotSupported to an OTA Requestor in the [QueryImageResponse](#) command if all the following conditions apply:

1. A new Software Image is determined to be available.
2. The Software Image to proxy is served by a remote server that does NOT support range-based transfers.
3. The OTA Requestor only supports BDX.
4. The OTA Provider does not support asynchronous proxying (e.g. download-then-proxy).

The fields of the [QueryImageResponse](#) command convey the next steps to take. The primary indication of action to be taken by the OTA Requestor is expressed in the Status field, with the other fields providing the necessary details as described in the following subsections.

Failure to receive an application-layer response from the OTA Provider after invoking the [QueryImage](#) command SHOULD be considered equivalent to having received a [QueryImageResponse](#) command with a Status field containing NotAvailable (see [Section 11.20.3.2.4, "Handling NotAvailable value in Status field"](#)).

Access Control Requirements

Commissioners or Administrators SHOULD install necessary [ACL entries](#) at Commissioning time or later to enable the handling of the [AnnounceOTAProvider](#) commands by OTA Requestors.

Below is an exemplary ACL entry for a Node implementing the OTA Requestor cluster server to support the processing of the [AnnounceOTAProvider](#) command:

```
{  
  FabricIndex: <Fabric Index of the fabric in question>,  
  ...  
}
```